$$u = \frac{2\phi_l}{\pi}, \quad 0 \leq \phi_l \leq \frac{\pi}{2}$$

$$p_l(\phi_l) = \begin{bmatrix} x_l^n \\ y_l^n \\ z_l^n + 0.2(-2u^3 + 3u^2) \end{bmatrix}$$